

Analytical Taylor series expansion of $g(\nu)$.

Taylor expansion of a function $f(\nu, \mu) = x \coth\left(\frac{x+\mu}{2}\right) - 4$ around $\mu = 0$ up to first order:

$$y(\hat{n}) \cdot f(\nu, \mu) \simeq y(\hat{n}) \cdot f(\nu, 0) + \mu \cdot y(\hat{n}) \cdot \left. \frac{\partial f(\nu, \mu)}{\partial \mu} \right|_{\mu=0} \quad (1)$$

where I will define

$$x \equiv \frac{h\nu}{k_B T_{CMB}} \quad (2)$$

so that

$$\left. \frac{\partial f(x, \mu)}{\partial \mu} \right|_{\mu=0} = x \cdot \left[-\frac{1}{\sinh^2\left(\frac{x+\mu}{2}\right)} \right]_{\mu=0} \cdot \frac{1}{2} = -\frac{x}{2} \frac{1}{\sinh^2\left(\frac{x}{2}\right)} \quad (3)$$

and in K_{CMB} units:

$$f(x, \mu) = \left[x \coth\left(\frac{x+\mu}{2}\right) - 4 \right] T_{CMB} \quad (4)$$

$$\left. \frac{\partial f(x, \mu)}{\partial \mu} \right|_{\mu=0} = \left[-\frac{x}{2} \frac{1}{\sinh^2\left(\frac{x}{2}\right)} \right] T_{CMB} \quad (5)$$

The final expression is the following one:

$$\boxed{y(\hat{n}) \cdot f(x, \mu) = y(\hat{n}) \cdot \left[x \coth\left(\frac{x+\mu}{2}\right) - 4 \right] \simeq y(\hat{n}) \cdot \left[x \coth\left(\frac{x}{2}\right) - 4 \right] \cdot T_{CMB} - \mu \cdot y(\hat{n}) \cdot \frac{x}{2} \frac{1}{\sinh^2\left(\frac{x}{2}\right)} \cdot T_{CMB}} \quad (6)$$